

Capping the Refinery, Not the Pump: Pass-Through of Korea's 2026 Wholesale Fuel Price Cap

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Abstract

When international fuel prices surged in 2026, Korea capped the prices refiners charge gas stations rather than the prices drivers pay, and revised the cap on announced dates. Using daily prices at about 10,700 stations and weekly refiner prices, I study the three decisions that changed the cap and seven that did not. On revision dates the share of stations changing price rose from 3–6% to 54–63%; on no-change dates it rose by at most 3.6 points. About 80% of each change in stations' supply cost reached pump prices within two weeks and about 90% within three. The June cut passed through faster where more rivals operated nearby: among same-brand stations in the same city, those with no rival within 1km had passed through 57% after one week and those with six or more 78%. A pre-cap pricing rule implies that the cap lay below refiners' cost-based price in every week, beyond the rule's error in 19–20 of 22, while the average retail margin did not change. Hypotheses were registered before estimation. Revisions of a wholesale cap reach consumers within weeks, more slowly in thin local markets; the gap between cost and cap falls on refiners and the budget.

Keywords: price cap; pass-through; retail fuel prices; market structure; price regulation; Korea

JEL codes: L71, Q41, L13, H22, D43

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1 Introduction

Price controls on fuel returned to the policy agenda in 2026. When the war in the Middle East and the disruption of shipping through the Strait of Hormuz pushed product prices to record levels, Korea introduced a maximum price on petroleum products for the first time since petroleum prices were liberalized in 1997 (Ministry of Trade and Industry, 2026e). Recent interventions elsewhere took two forms. Germany and other European countries cut fuel taxes temporarily in 2022, and their pass-through to pump prices has been studied in detail (Dovern et al., 2023; Drolsbach et al., 2023; Kahl, 2024). Hungary capped retail prices directly in 2021–22, and prices rose well above a synthetic control after the cap was lifted (Berezvai and Helfrich, 2024). Korea did neither. It capped the price at which the four refiners sell gasoline, diesel and kerosene to stations and distributors, left retail prices free, and revised the cap every two weeks at first and roughly every four weeks from June, announcing each new level on the evening before it took effect.

A wholesale cap raises three questions that matter for its design and that dominated the Korean debate. First, does a cap on refiners’ prices reach drivers, and how fast? Stations remain free to set prices, so the stated aim of protecting consumers depends on how they respond. Second, where does it reach drivers last? If pass-through is slower in thin local markets, the same national policy delivers less to drivers with few nearby alternatives. Third, did the cap actually bind, and who bore its cost? By September 2026 refiners were seeking compensation for losses they put at four to five trillion won, the government had promised to compensate them on a cost basis through a settlement committee (Ministry of Trade and Industry, 2026d), and no estimate of the gap weighted by sales volumes had been published.

This paper answers these questions with daily prices from about 10,700 Korean stations between December 2024 and September 2026, weekly refiner supply prices before and after tax, Singapore spot product prices, exchange rates and national shipment volumes. The cap was set or revised ten times. Three decisions changed its level: its introduction on 13 March, a 210-won increase on 27 March and a 150-won cut on 27 June. Seven left it unchanged. Because each revision moved every station’s cost by a known amount on a known date, the revisions provide a clean setting for measuring pass-through, and the seven no-change decisions provide placebo dates. I registered the hypotheses, estimators and decision rules before estimation and report every deviation from them.

Four findings stand out. First, revisions concentrated price changes on the revision date. The share of stations changing price rose from 3–6% on ordinary days to 54–63% at refiner-brand stations when the cap was introduced, while no comparable spike occurred on any of the seven no-change dates. Second, adjustment was fast but gradual. About 80% of each change in stations’ supply cost reached pump prices within two weeks and about 90% within three, with first price changes typically within zero to two days. Third, the cut passed through faster where more rivals operated nearby. Within a brand and province, stations in cities with more stations passed through more of the cut in its first week. Within the same city and brand, stations with more rivals within 1 km did so as well, and the gradient is large. Fourth, a pricing rule estimated on the pre-cap period implies that the cap lay below the price refiners would have charged in every week of the cap period, and

beyond the rule’s error in 19 of 22 weeks for gasoline and 20 for diesel; refiners priced within a few won of the cap throughout, and the average gap between retail prices and refiners’ supply prices did not change. The gap between cost-based prices and the cap, an opportunity cost rather than an accounting loss, fell on refiners and, through the concurrent fuel-tax cut, on the public budget.

These are pass-through rates of observed changes in the regulated supply price, measured from the price on the day before each revision. They answer how fast and where a revision reaches drivers; they do not measure how far the cap held prices below their level without it, a question for which no domestic control group exists.

1.1 Related work

The paper speaks to three strands of work. The first studies how temporary fuel-price interventions reach consumers. Germany’s three-month fuel-tax cut in 2022 was passed through largely and quickly: Dovern et al. (2023) find, with a synthetic difference-in-differences estimator on weekly national prices, that the cut was passed through fully for most of the period. Drolsbach et al. (2023), using daily chain-level prices and a staggered difference-in-differences design with untreated countries as controls, find full shifting of the 2022 tax cuts in France, Germany and Italy, no significant change in average retail margins, and a few days’ delay before full pass-through. Kahl (2024), comparing German and French stations daily, estimates average pass-through of 87–91%, with large swings over time and lower rates in rural than in urban areas (82% against 92%). For Korea, Jang (2021) finds that 87% of the 2018–19 gasoline tax cut and 73% of the diesel cut reached prices at 541 Seoul stations, more at self-service and Altteul stations, while Jang and Choi (2023), using province averages, estimate much lower pass-through of the 2021–22 cut. Hungary’s retail cap of 2021–22 is the closest analogue to a price ceiling; Berezhvai and Helfrich (2024) find that prices exceeded a synthetic control by 12% in the ten months after its removal. None of these studies a cap on wholesale prices. I show that revisions of a cap aimed at refiners reach pump prices about as completely as a tax cut, but over two to three weeks.

The second strand studies the speed of retail fuel price adjustment and its relation to market structure. Since Borenstein et al. (1997), the literature has asked whether retail prices rise faster than they fall; Korean studies reach mixed conclusions depending on the cost measure and period (Kim and Kim, 2009; Ko, 2013; Kim, 2017, 2022). Evidence on whether local competition matters for adjustment is also mixed. Valadkhani et al. (2021) find that the 672 stations of Greater Sydney passed through less of wholesale and crude oil price changes when fewer than 13 rivals operated within 5 km, and propose this threshold as a rule of thumb for monitoring. Deltas (2008) finds that U.S. states with high retail margins passed through 60% of a cost change within a month against 70% in low-margin states, with greater asymmetry; Verlinda (2008) finds in Southern California that each additional rival within a tenth of a mile reduces the asymmetry of adjustment; and Hong and Lee (2020) find that Korean stations on isolated islands raised prices within about five weeks of a cost increase but took more than ten weeks to adjust to a decrease, which they attribute to tacit collusion. But Lewis and Noel (2011), measuring across 72 U.S. cities the share of a wholesale cost

shock passed through after a given number of days, find that the presence of Edgeworth price cycles, not station density, governs speed: cities without cycles pass through about 75% of an increase after a week and take about six weeks to complete a decrease, while cities with cycles pass through within five to nine days. In Korea, Jang (2021) finds no interaction between the number of stations within 1 km and the tax-cut period, and other evidence on local competition concerns price levels rather than adjustment (Jung et al., 2013; Hong, 2017). Closest to this paper, Kim and Ni (2022) study daily prices at 720 Seoul stations in 2009–2019 and find that cost decreases were passed through more fully than increases in the more competitive segments of the market—self-service stations and high-volume districts—and that the 2018–19 tax cut reached prices faster than its later reversal; they measure competition by station type and district rather than by nearby rivals, and they study ordinary cost shocks rather than a regulated wholesale price. Korean stations do not cycle. Using within-city variation across 10,000 stations and a cost shock that is identical for all of them and known in advance, I find that pass-through was faster where more rivals operated nearby, though the level reached after three weeks did not differ. To my knowledge this is the first nationwide station-level evidence on how the pass-through of a binding wholesale price cap varies with local market structure.

The third strand studies price ceilings as focal points. Knittel and Stango (2003) show that state ceilings on credit-card interest rates served as focal points for tacit collusion, identified from variation in ceilings across states and over time and from periods when the ceiling did not bind. Zhang et al. (2020) find that the modal price of stations in a Chinese city coincided with the government ceiling and that stations jumped to the ceiling as their prices approached it, and Cui et al. (2023), using daily prices at 2,819 stations in Inner Mongolia, find that retail prices followed changes in China’s retail ceiling rather than crude oil prices, rising quickly when the ceiling was raised and falling slowly when it was lowered. These ceilings apply to retail prices, and the focal-point tests require periods or places in which the ceiling exceeds the competitive price. I show that the Korean cap never exceeded refiners’ cost-based price by more than the pricing rule’s error, so that refiners’ parallel prices just below it carry no information on coordination, and I document that a standard regression test for timing coordination among stations is biased toward finding it.

Section 2 describes the policy and the data. Section 3 sets out the empirical approach and Section 4 the results, including robustness. Section 5 discusses why local competition matters and what the results imply for policy. Section 6 concludes.

2 The 2026 maximum price and the data

2.1 The policy

On 12 March 2026 the Ministry of Trade and Industry announced maximum supply prices for regular gasoline, diesel and kerosene sold by the four refiners—SK Energy, GS Caltex, HD Hyundai Oilbank and S-OIL—to stations and distributors, effective the next day, under a ministerial notice on maximum petroleum prices and the restriction of excessive exports (Ministry of Trade and

Table 1: Cap decisions and fuel-tax schedule, 2026

Decision	Effective date(s)	Gasoline	Diesel	Kerosene	Change
1st	13 Mar	1,724	1,713	1,320	Introduction
2nd	27 Mar	1,934	1,923	1,530	+210 (fuel tax cut widened)
3rd–6th	10 Apr, 24 Apr, 8 May, 22 May	1,934	1,923	1,530	Unchanged
7th	27 Jun	1,784	1,773	1,380	–150
8th–10th	25 Jul, 22 Aug, 19 Sep	1,784	1,773	1,380	Unchanged
Fuel tax incl. VAT, gasoline / diesel (won/L): 698 / 448 (Nov 2024–Apr 2025); 739 / 495 (May–Oct 2025); 764 / 524 (Nov 2025–26 Mar 2026); 698 / 436 (from 27 Mar 2026).					

Caps in won per liter, inclusive of taxes. Each decision was announced on the evening before it took effect. Tax amounts are implied by Opinet’s weekly after-tax and pre-tax refiner prices (after-tax minus 1.1 times pre-tax).

Industry, 2026e,c). Retail prices and premium gasoline were not covered. The first caps were 1,724 won per liter for gasoline, 1,713 won for diesel and 1,320 won for kerosene, inclusive of taxes. The cap was defined by a formula: a pre-tax base price, initially refiners’ average supply price before the Middle East crisis and thereafter the previous pre-tax cap, multiplied by the recent change in Singapore product prices, plus taxes (Ministry of Trade and Industry, 2026e). To prevent refiners from diverting supply abroad, exports of capped products were limited in principle to their volume in the same period of 2025. Refiners’ losses were to be compensated from the budget under Article 23(3) of the Petroleum Business Act, settled quarterly by a settlement committee on the basis of costs; the government declined to put a figure on them (Ministry of Trade and Industry, 2026f,d).

Table 1 lists the decisions. The cap rose by 210 won on 27 March, was left unchanged on 10 April, 24 April, 8 May and 22 May, fell by 150 won on 27 June, and was left unchanged again on 25 July, 22 August and 19 September. Each decision was announced on the evening before it took effect. The government departed from its own formula in both directions. When it froze the cap on 24 April, it stated that the formula alone implied cuts of about 100 won for gasoline and 200 won for diesel, but that increases left unreflected in earlier rounds amounted to about 125 and 628 won (Ministry of Trade and Industry, 2026a). The cap therefore sat well below the formula price during the spring.

A temporary fuel-tax cut ran throughout. Its rate was narrowed in November 2025 to 7% for gasoline and 10% for diesel and widened on 27 March 2026 to 15% and 25%, the same day as the cap increase, lowering the tax including VAT by 65 won per liter for gasoline and 87 won for diesel (Ministry of Economy and Finance, 2025). Because the cap and refiners’ reported supply prices both include taxes, the 210-won increase is the net change in stations’ after-tax cost. I verified the tax schedule week by week from the difference between Opinet’s after-tax and pre-tax supply prices; press summaries had dated the November 2025 change to January 2026.

About 10,700 stations sell one of the four refiner brands or the discount Altteul brand. Altteul stations, introduced in 2011, buy fuel through a weekly joint purchase run by the Korea National Oil Corporation (Jung et al., 2013; Hong, 2017). Opinet, the government’s price portal, publishes each station’s prices daily and refiners’ average supply prices weekly. Korea has intervened at the

wholesale level once before, when refiners cut supply prices temporarily in April–July 2011 (see Kim, 2017); no legal ceiling was involved and the cut was not revised on announced dates.

2.2 Data

Station prices. Opinet publishes the daily price of each station for regular gasoline, premium gasoline, diesel and kerosene. I use all stations from 29 December 2024 to 30 September 2026 (641 days) and keep the four refiner brands and Altteul, including its highway and agricultural-cooperative variants: 10,744 stations. Each station’s city (*si-gun-gu*, 229 units) is fixed at its modal label before 1 July 2026, when the merger of Gwangju and South Jeolla and the reorganization of Incheon’s districts changed labels. A price that moves by 300 won or more and returns the next day is treated as a recording error (147 gasoline and 238 diesel observations).

Locations. I geocoded station addresses with the VWorld geocoder of the Ministry of Land, Infrastructure and Transport and obtained coordinates for 10,795 of 10,885 stations (99.2%), correcting addresses that still carried pre-reform township names. For each station I count rivals within 1 km among stations active in February 2026.

Refiner prices, costs and volumes. Opinet reports weekly average supply prices of all refiners and of each refiner, after and before tax. Product costs are Singapore spot prices for 92 RON gasoline and 0.001% sulfur diesel converted at the Bank of Korea’s daily won–dollar rate. National monthly shipments by product and province come from the Korea National Oil Corporation’s Petronet.

Table 2 describes the 10,313 stations active in February 2026. Sixty-one percent are self-service. The median station lies in a city with 65 stations and has two rivals within 1 km on average; 28% have none. Altteul stations are more often on highways and less often near rivals. Before the cap a station changed its gasoline price on 9% of days; during the cap, on 5.5%. Korean stations do not show Edgeworth cycles: the median daily change of city-average prices is zero in almost every city, and no city meets the cycling criterion of Lewis and Noel (2011).

3 Empirical approach

Pass-through paths. For a revision on day τ that changes the cap by Δc , station i ’s cumulative pass-through after r days is

$$PT_{i,r} = \frac{p_{i,\tau+r} - p_{i,\tau-1}}{\Delta c}. \quad (1)$$

I use the increase ($\Delta c = +210$) and the cut ($\Delta c = -150$). The introduction on 13 March is used for timing only: refiners’ weekly prices straddle the introduction date, so its cost change is not observed precisely. Each event sample keeps stations observed on at least 80% of days from a week before to three weeks after the revision, observed the day before, and not changing brand. Confidence intervals cluster by city.

Table 2: Stations active in February 2026

	SK	GS	HD	S-OIL	Altteul	All
Stations	2,589	1,962	2,212	2,240	1,310	10,313
Self-service (%)	58.5	63.8	58.3	66.8	59.0	61.3
Highway (%)	0.2	0.1	0.0	0.1	16.0	2.1
Stations in city, median	65	68	70	66	59	65
Rivals within 1 km, mean	2.0	2.1	2.1	2.0	1.4	2.0
No rival within 1 km (%)	26.7	26.0	27.4	27.9	36.0	28.2
Gasoline price, Feb 2026 (won/L)	1,697	1,694	1,690	1,689	1,663	1,689
Gasoline price, Apr–Jun 2026 (won/L)	2,014	2,011	2,009	2,010	1,993	2,009
Days with a price change, Jan 2025–Feb 2026 (%)	8.1	8.6	9.2	8.5	11.5	9.0
Days with a price change, cap period (%)	5.2	5.2	5.7	5.5	6.2	5.5

SK = SK Energy, GS = GS Caltex, HD = HD Hyundai Oilbank. Stations with a gasoline price in February 2026. The cap period here runs from 12 April to 30 September 2026, after the adjustment to the second decision.

Market structure. I relate pass-through after r days to local competition,

$$PT_{i,r} = \beta \log N_i + \gamma \text{Self}_i + \delta \text{Highway}_i + \alpha_{b(i),g(i)} + u_i, \quad (2)$$

in two versions. In the registered version N_i is the number of stations in station i 's city and $\alpha_{b,g}$ a brand-by-province effect. In the within-city version N_i is one plus the number of rivals within 1 km and $\alpha_{b,g}$ a brand-by-city effect, so that only variation across stations of the same brand in the same city is used. Standard errors cluster by city (229 clusters), and I report wild cluster bootstrap p -values (Cameron et al., 2008) with 999 Rademacher draws. The seven no-change decisions serve as placebos.

Timing coordination. The registered test of coordination asks whether, after a revision, two stations of the same brand in the same city changed price on the same day more often than two stations of different brands. A standard regression of the same-day indicator on a same-brand dummy with city-by-day and brand-by-day effects is biased toward a positive coefficient even when stations move independently: if brand b changes price on a given day with probability p_b , independent same-day changes occur with probability $p_b p_{b'}$ for a cross-brand pair but $(p_b^2 + p_{b'}^2)/2$ on average for same-brand pairs, which is larger whenever $p_b \neq p_{b'}$. I therefore use as the dependent variable the excess of the same-day indicator over the product of the two stations' brand-province-day change rates, computed leaving the pair out. This correction was fixed before estimation; I report the standard regression alongside it.

Attribution to the cap. Price changes in an event window could reflect international prices rather than the cap. International prices are common to all stations and cannot be separated from event-time effects directly. I therefore test the premise that, while the cap binds, international prices do not move retail prices. I regress the daily change in the average retail price of refiner-brand stations on the current and 14 lagged daily changes in Singapore product costs, with day-of-week effects and Newey–West standard errors, separately before the cap and during cap periods away

from revisions.

Was the cap binding? I estimate, on the 60 weeks before the cap, a rule relating refiners’ average pre-tax supply price to the current and two lagged weekly product costs, and apply it to the cap period to obtain the price refiners would have charged without the cap. A week is classified as binding if the cap lies more than two residual standard deviations below this price, and as a potential floor—the case studied by Knittel and Stango (2003)—if it lies more than two above; weeks in which the gap is within two standard deviations are left unclassified.

Registration. The hypotheses on market structure, timing coordination and price dispersion, the estimators, the placebo rule and the decision criteria were frozen with cryptographic hashes before estimation, as were the binding test and the robustness analyses. The registered test of timing coordination and the registered test of a decline in between-brand price dispersion failed and are reported in Sections 4.5 and 4.6. The asymmetry between the increase and the cut was noticed after part of the data had been seen and is reported as descriptive.

4 Results

4.1 Price changes concentrate on revision dates

Figure 1 shows the daily share of stations changing their gasoline price. On ordinary days in 2026, 3–6% of stations changed price. On 13 March, 54–63% of refiner-brand stations and 89% of Altteul stations did. About 40% changed on the day after the March increase and on the Monday after the Saturday cut in June. On the seven no-change dates, the share within two days of the decision exceeded its average over the preceding week by at most 3.6 percentage points, against 29–35 points on revision dates. The busiest day of 2026 preceded the cap: in early March, during the surge in international prices, 83% of refiner-brand stations changed price on a single day.

Prices did not move ahead of the increase or the cut. On the announcement days of 26 March and 26 June, mean prices moved by less than two won per liter, as on the preceding five days. Only the introduction, announced on 12 March, saw prices fall by 6–10 won on the announcement day.

4.2 About 80% within two weeks

Figure 2 shows pass-through paths with 95% confidence intervals. The paths measure the response of pump prices to the change in stations’ after-tax supply cost at each revision; for the March event that cost change combines the cap increase with the widening of the fuel-tax cut, and the 210 won is the net amount. For refiner brands, 53% of the gasoline increase had reached pump prices after one week, 80% after two and 86% after three; for the cut, 68%, 84% and 88%. Diesel paths are similar (50%, 79% and 85% for the increase; 70%, 87% and 93% for the cut). Altteul stations followed a different path: they barely moved in the first two days of the increase, jumped on the third and then overtook refiner brands, reaching 89% after three weeks, while they lagged slightly

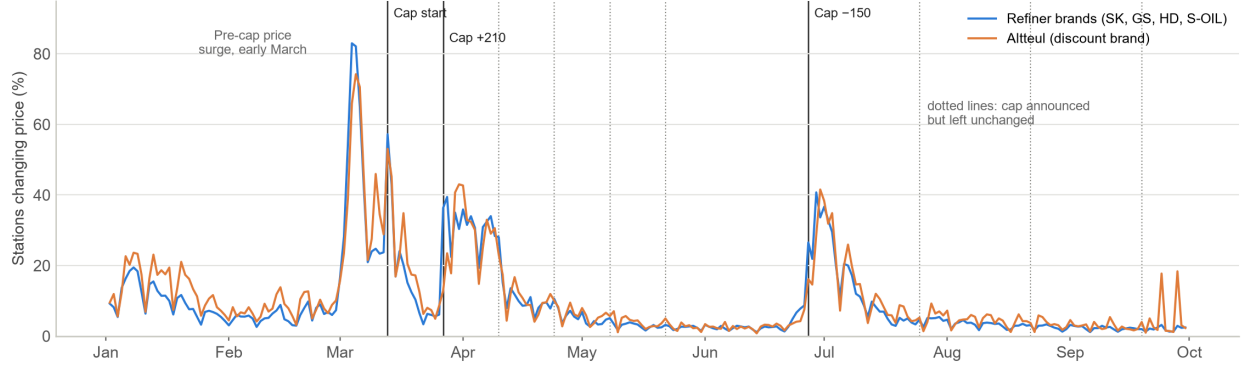


Figure 1: Share of stations changing their gasoline price each day, 2026. Solid lines mark cap decisions that changed the cap; dotted lines mark decisions that left it unchanged. The early-March spike precedes the cap and coincides with the surge in international prices.

after the cut. This pattern fits weekly procurement through the joint purchase. Figure 3 contrasts the revision paths with the seven no-change dates, on which mean prices moved by a few won; the exception, 10 April, falls within the adjustment to the 27 March increase.

The two events differ in speed. Comparing the same stations in both event samples, pass-through one week after the increase was 13 percentage points below that after the cut for gasoline (cluster $t = -13.1$, wild bootstrap $p = 0.001$) and 18 points for diesel; after two weeks the gap was 2 and 7 points. This is the reverse of the familiar asymmetry (Borenstein et al., 1997). It agrees with Kim and Ni (2022), who find that Korea’s 2018–19 tax cut reached prices faster than its reversal and that competitive segments pass through decreases more fully, but it runs against Kim (2022), who finds that refiner-to-station increases pass through faster when prices are high, and against Cui et al. (2023), who find that Chinese stations followed increases in a retail ceiling faster than decreases. Under a retail ceiling the cap itself is the price many stations charge; under a wholesale cap it is a cost, and stations have no focal retail price to hold. The finding rests on one event of each kind, the increase coincided with a larger fuel-tax cut that softened the price change drivers saw, and fuel prices were under close government and press scrutiny. I report it as descriptive.

4.3 Faster where more rivals operate nearby

Table 3 reports the market-structure regressions one week after each revision. In the registered city-level specification, a city with twice as many stations had passed through 3.7 percentage points more of the June cut after one week ($\hat{\beta} = 0.054$, s.e. 0.010), about 5.6 won. For the March increase, city size did not matter in the first week (-0.006 , s.e. 0.009) and mattered by the second (0.023, s.e. 0.006). The registered pooled estimate over the three revisions is 0.053 (s.e. 0.016) for gasoline and 0.039 (0.010) for diesel; it also uses the introduction, with pass-through scaled by refiners’ weekly price change, and is a different estimator from the two-event pooled column of Table 3. On the no-change dates the same coefficient on the one-week price change is -0.34 won, less than a

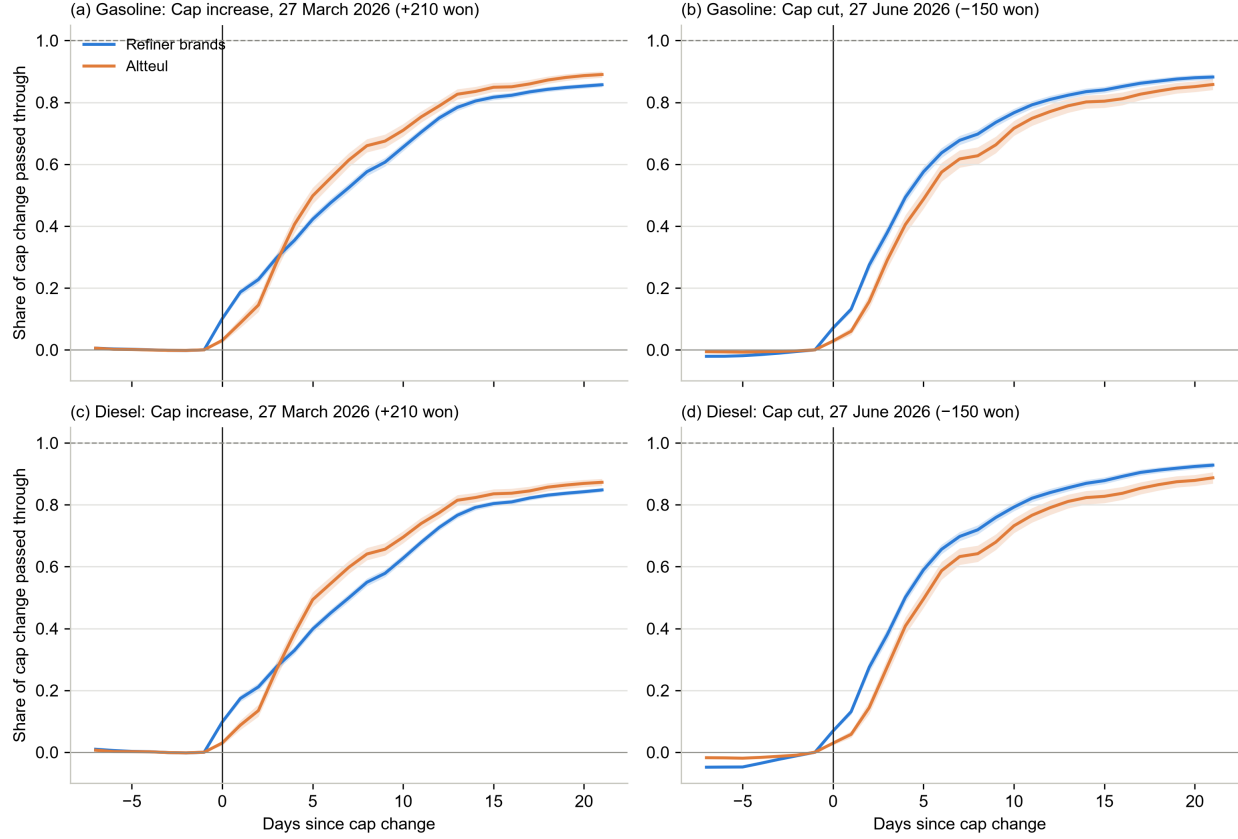


Figure 2: Cumulative pass-through of the cap increase (27 March) and the cap cut (27 June), by brand group, with 95% confidence intervals clustered by city. Pass-through is the change in a station’s price since the day before the revision divided by the change in the cap. The dashed line marks full pass-through.

twentieth of the effect on revision dates. Self-service stations passed through 12.7 points more of the cut after one week and highway stations 15.4 points more.

The within-city measure gives a sharper result. Comparing stations of the same brand in the same city, a one-unit increase in the log of one plus the number of rivals within 1 km raised one-week pass-through of the cut by 4.2 percentage points for gasoline and 3.8 for diesel, and of the increase by 2.1 and 1.1 points. One week after the cut, stations with no rival within 1 km had passed through 57% of it, those with one or two rivals 68%, three to five 74%, and six or more 78%. The effect fades over the following weeks: by the second and third week, pass-through converges across market structures. Figure 4 shows the geography. Stations without a rival within 1 km, 28% of the total, line the roads of the rural interior, while dense clusters lie in Seoul and the larger cities. One week after the June cut, pass-through was highest around the Seoul metropolitan area and the large cities of the southeast and lowest in rural areas: across hexagonal cells with at least five stations, the 10th, 50th and 90th percentiles of mean pass-through were 44%, 65% and 80%.

These results agree in direction with Deltas (2008), Verlinda (2008), Hong and Lee (2020), Valadkhani et al. (2021) and the rural–urban gap in Kahl (2024). They differ in one respect: in

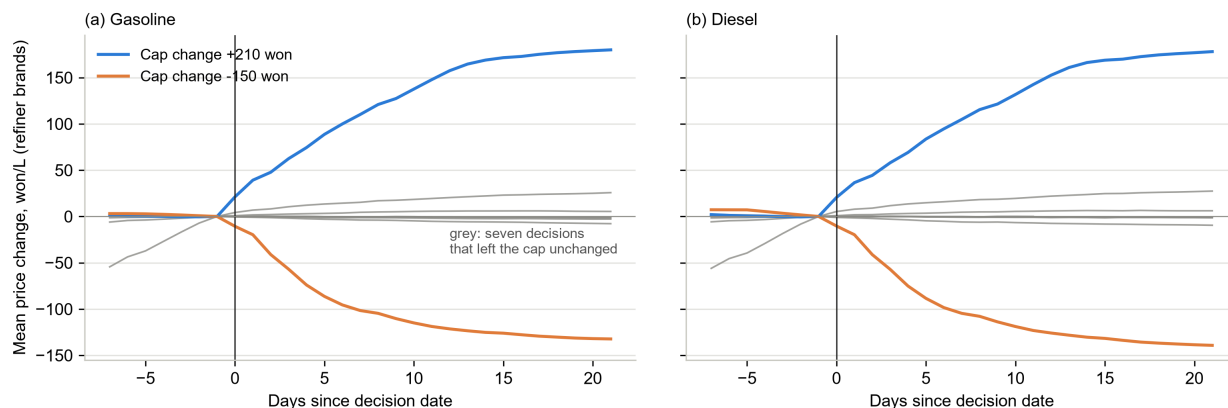


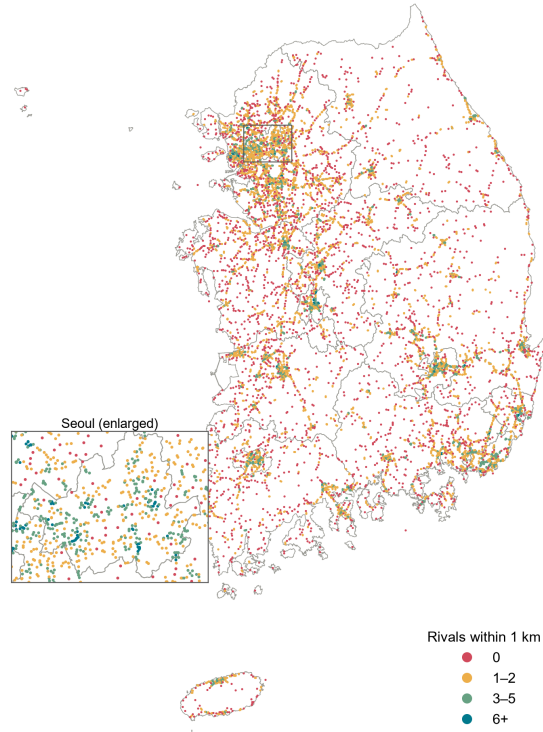
Figure 3: Mean price change of refiner-brand stations around cap decisions, in won per liter. Grey lines are the seven decisions that left the cap unchanged; the rising grey line is 10 April, within the adjustment to the 27 March increase.

Sydney and Germany stations with fewer rivals passed through less over the whole period studied, whereas in Korea the gap closed within three weeks. A one-off, publicly announced cost change of known size may leave less room for persistent differences than the continuous cost movements those studies examine. The results also differ from two other findings in ways that the measures explain. Jang (2021) found that the number of stations within 1 km did not change how much of a tax cut stations passed through over several months; the association I find concerns speed in the first week and has largely disappeared by the third. Lewis and Noel (2011) found that across U.S. cities station density is unrelated to the speed of pass-through once Edgeworth cycles are accounted for; Korean stations do not cycle, and the within-city comparison removes city-level differences that a cross-city regression cannot.

4.4 Where the cap stood relative to cost, and who bore the gap

During the cap period, away from revisions, international prices did not move retail prices. Before the cap, 31% of a change in Singapore product costs reached average retail prices within 15 days (gasoline 0.312, s.e. 0.064; diesel 0.306, s.e. 0.064). During cap periods away from revisions the cumulative response was -0.016 (0.011) for gasoline and -0.014 (0.002) for diesel. The diesel estimate is statistically distinguishable from zero—so the registered criterion is formally not met—but it is a twentieth of its pre-cap size and has the wrong sign; it does not indicate that international prices pushed diesel prices up. The pass-through estimates therefore measure the response to the revisions rather than to concurrent movements in international prices, with the caveat that the diesel criterion was formally missed. The magnitudes show that the caveat is minor. Over the three weeks after the March increase, the Singapore diesel cost rose by 760 won per liter in the first week and ended 460 won below its starting level; at the pre-cap response of 0.31 such swings would have moved retail diesel prices by about +230 and then -140 won, whereas at the cap-period estimate they move them by -11 to $+8$ won, against observed changes of $+105$ and $+178$ won that

(a) Rival stations within 1 km, February 2026



(b) Share of the 27 June cut passed through after one week

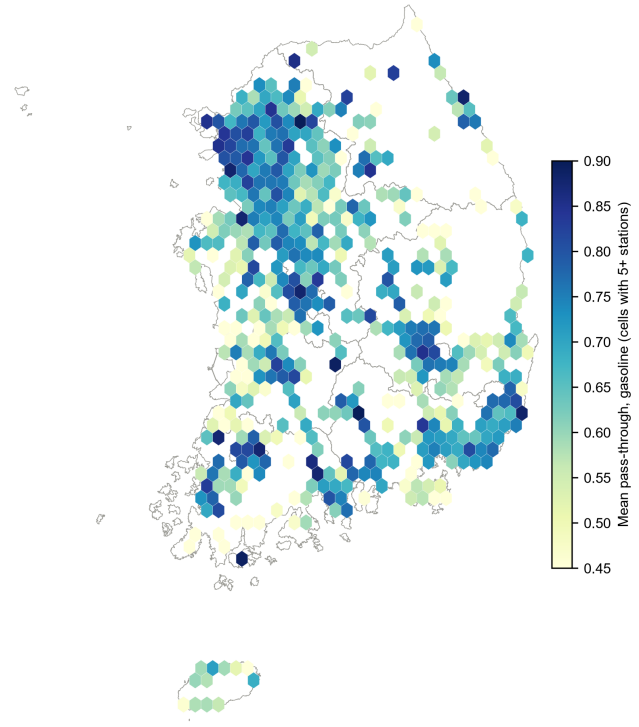


Figure 4: Local competition and pass-through of the 27 June cut. Panel (a): stations active in February 2026 by the number of rival stations within 1 km, with an enlarged view of Seoul. Panel (b): mean one-week pass-through of the cut for gasoline, averaged over stations in hexagonal cells with at least five stations. Province boundaries from Statistics Korea (2018); two stations on Ulleung Island are not shown.

track the 210-won revision. In the seven no-change windows, diesel costs moved by between -300 and $+180$ won per liter over three weeks while refiner-brand diesel prices moved by at most 27 won, and by at most 10 won outside the 10 April window. The registered criterion is a significance test; with retail prices nearly constant during the cap, the standard error is small enough to reject a response of 1.4 percent, which is why the statistical verdict and the economic conclusion differ. I keep the registered verdict and report the magnitudes alongside it.

Figure 5 compares refiners' supply prices with the cap and with the price implied by the pre-cap rule (adjusted R^2 of 0.92 for gasoline and 0.95 for diesel; residual standard deviation of 9–10 won before tax). In every one of the 22 cap-period weeks, excluding transition weeks, the cap lay below the implied price. In 19 weeks for gasoline and 20 for diesel the gap exceeded twice the rule's residual standard deviation, the registered criterion for a binding week; in the remaining weeks, in mid-June and early July when product prices had fallen, the gap was within the rule's error and the test cannot tell whether the cap bound. In no week did the cap exceed the implied price by more than the error, so there is no week in which it could have acted as a floor. The gap reached about 200 won per liter for gasoline and 1,000 won for diesel before tax, largest in early April and,

Table 3: Local competition and one-week pass-through

	Stations in city (log)			Rivals within 1 km, $\log(1+n)$		
	Increase	Cut	Pooled	Increase	Cut	Pooled
<i>A. Gasoline</i>						
Coefficient	−0.006 (0.009)	0.054 (0.010)	0.024 (0.008)	0.021 (0.006)	0.042 (0.006)	0.032 (0.004)
Wild bootstrap p	0.594	0.001	0.002	0.001	0.001	0.001
Stations	10,148	10,032	20,180	10,097	9,984	20,081
<i>B. Diesel</i>						
Coefficient	−0.006 (0.010)	0.057 (0.010)	0.026 (0.008)	0.011 (0.005)	0.038 (0.006)	0.025 (0.004)
Wild bootstrap p	0.611	0.001	0.005	0.037	0.001	0.001
Stations	10,188	10,074	20,262	10,136	10,026	20,162
Fixed effects	Brand $\times$ province (pooled: $\times$ event)			Brand $\times$ city (pooled: $\times$ event)		

Dependent variable: share of the cap change passed through seven days after the revision. Increase = 27 March (+210 won); cut = 27 June (−150 won); pooled = these two events only (the registered three-event estimate in the text also includes the 13 March introduction and is a different estimator). All regressions include self-service and highway indicators. Standard errors clustered by city (229 clusters) in parentheses; wild cluster bootstrap with 999 Rademacher draws.

for gasoline, again in mid-September. Crude oil prices alone would suggest that the cap exceeded cost in July and August, but product cracks widened sharply in 2026 and on a product basis the cap never did. Refiners’ average prices sat 1–7 won below the cap in every week but one, which is what a ceiling at or below cost produces under any form of competition. Parallel pricing at the cap therefore carries no information on coordination, and a focal-point test in the spirit of Knittel and Stango (2003) cannot be run.

The average retail margin did not change. The measure is the national average retail price of refiner-brand stations minus refiners’ average after-tax supply price; Opinet does not report individual stations’ purchase prices, so it describes stations as a group rather than any station’s margin over its own cost. For gasoline, this gap averaged 71 won in 2025, 89 won in January–February 2026, 77 won in April–June and 87 won in July–September; for diesel, 81, 89, 82 and 83 won. Drolsbach et al. (2023) likewise found no change in average retail margins after the European tax cuts. Weighting the gap between implied and actual supply prices by national shipments gives about 3.0 trillion won for April–August, against about 2.0 trillion won of revenue forgone through the fuel-tax cut over the same months. The first figure values refiners’ sales at international product prices; it is an opportunity cost, not an accounting loss, and it relies on the pre-cap rule holding at the extreme prices of 2026. It is neither refiners’ reported loss nor the compensation the settlement committee will determine, which had not been set by September 2026 and is the cap’s direct fiscal cost. The second figure is the budget cost of the tax cut, a separate instrument, and is shown for scale.

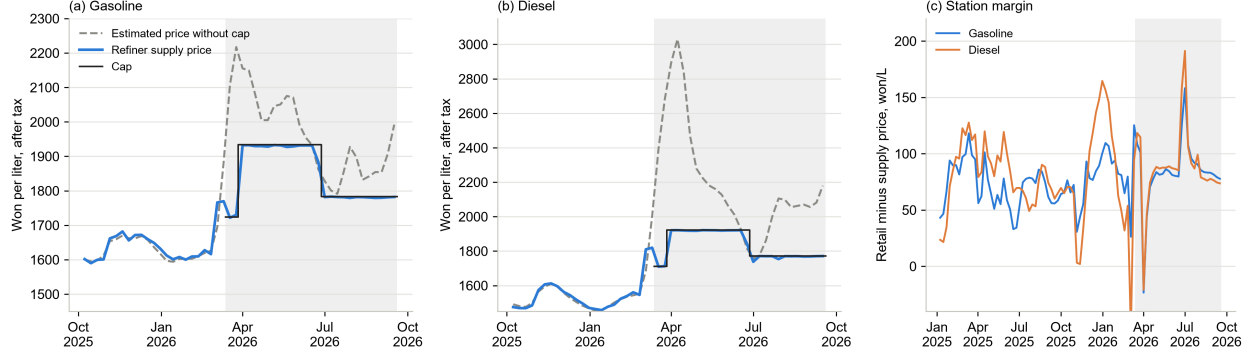


Figure 5: Refiners’ weekly average supply price, the cap, and the price implied by the pre-cap rule, after tax (panels a and b); station margins, retail price of refiner-brand stations minus refiners’ supply price (panel c). Shading marks the cap period. Spikes in margins in transition weeks reflect weekly averages that straddle the revision date.

Table 4: Same-day price changes of same-brand pairs, per 1,000 pair-days

	Gasoline		Diesel	
	Revisions	No-change dates	Revisions	No-change dates
Standard additive regression	2.80 (0.63)	0.94 (0.11)	3.05 (0.66)	0.85 (0.12)
Excess over independent timing	−0.89 (0.64)	0.17 (0.11)	−0.63 (0.67)	0.02 (0.12)
Excess, Altteul pairs, revisions	8.60 (2.77)		7.52 (2.84)	

Eight days after each decision. Revisions: 13 March, 27 March, 27 June; no-change dates: the seven decisions that left the cap unchanged. Standard errors clustered by city in parentheses.

4.5 No timing coordination among same-brand stations

Table 4 reports the coordination test on pairs of stations in the same city (about 178,000 pairs and 1.4 million pair-days per event) over the eight days after each decision. The standard regression finds same-brand “coordination” on revision dates and, at a third of the size, on dates when nothing happened: the bias described in Section 3 is at work. Once the independence benchmark is removed, same-brand pairs of refiner brands show no excess same-day changes, on revision dates or otherwise. Altteul pairs are the exception, with 8–9 excess same-day changes per 1,000 pair-days after revisions, consistent with their weekly joint procurement. The data therefore give no sign that neighboring stations of the same brand timed their changes together; the clustering of changes on revision dates reflects a common, public cost change.

4.6 Robustness

Table 5 varies the specification of the pooled city-level regression and the within-city regression. Requiring 70% or 90% of days observed, or dropping highway stations, leaves the city-level coefficient between 0.024 and 0.027. Brand by brand, the coefficient is positive for every brand and largest for Altteul; it is not significant for HD Hyundai Oilbank. Dropping the 112 stations located by place-name search leaves the within-city estimates unchanged, and replacing brand-by-city

Table 5: Robustness of the competition estimates

	Gasoline	Diesel
<i>Stations in city (log), increase and cut pooled</i>		
Baseline (80% of days observed)	0.024 (0.008)	0.026 (0.008)
70% of days observed	0.024 (0.008)	0.026 (0.008)
90% of days observed	0.024 (0.008)	0.025 (0.008)
Excluding highway stations	0.025 (0.008)	0.027 (0.008)
SK Energy	0.020 (0.009)	0.025 (0.009)
GS Caltex	0.030 (0.010)	0.036 (0.010)
HD Hyundai Oilbank	0.013 (0.009)	0.012 (0.009)
S-OIL	0.019 (0.010)	0.022 (0.011)
Altteul	0.049 (0.013)	0.041 (0.012)
Controlling for pre-cap change frequency	0.021 (0.007)	0.022 (0.008)
<i>Rivals within 1 km, log(1+n)</i>		
Cut, excluding place-search coordinates	0.042 (0.006)	0.039 (0.006)
Increase, excluding place-search coordinates	0.021 (0.006)	0.011 (0.005)
Cut, brand $\times$ province effects	0.065 (0.006)	0.061 (0.006)
Increase, brand $\times$ province effects	0.020 (0.007)	0.007 (0.006)
Cut, controlling for pre-cap change frequency	0.035 (0.006)	0.034 (0.006)
Increase, controlling for pre-cap change frequency	0.018 (0.006)	0.010 (0.005)

Dependent variable: one-week pass-through. Standard errors clustered by city in parentheses. Pre-cap change frequency: share of days in January 2025–February 2026 on which the station changed its price; these rows were added after review.

with brand-by-province effects makes them larger for the cut. Re-estimating the binding test with after-tax prices and the verified tax schedule gives the same classification.

A check added after review controls for each station’s pre-cap frequency of price changes (the share of days from January 2025 to February 2026 on which it changed its price), which could proxy for turnover or pricing habits. The frequency itself predicts pass-through: for the gasoline cut, a station that changed its price on ten more days per hundred before the cap passed through about 7 percentage points more after one week. The coefficient on rivals within 1 km falls by about a sixth and remains (0.035 for the cut and 0.018 for the increase for gasoline; 0.034 and 0.010 for diesel), and the city-level coefficient falls from 0.024 to 0.021. The correlation between pre-cap frequency and the number of rivals within 1 km is 0.19 for gasoline.

Price dispersion. The registered hypothesis that the cap compressed price differences between brands within a city is not supported. Within-city dispersion fell by 40–60% in both its between-brand and within-brand components, but before the cap dispersion was already lower when price levels were higher, and the decline lies within what that relation predicts.

Diesel shipments. National diesel shipments fell 12.1% year on year in March–August 2026, against an average decline of 5.3% in 2025, while gasoline shipments fell 3.5% against a 2.3% rise. The government attributes the larger diesel decline to the shrinking diesel fleet, 17% smaller than in 2020 (Ministry of Trade and Industry, 2026b). Higher diesel prices (31% above a year earlier in April, against 20% for gasoline) and weaker freight activity in August are also consistent with it. Export limits would, if anything, have raised domestic shipments. I do not attribute the decline to

the cap.

5 Discussion and policy implications

5.1 Why was pass-through faster where rivals were closer?

Three explanations are consistent with the pattern that the cut reached prices faster where rivals were closer, that the increase did so only weakly, and that the difference disappeared within three weeks.

The first is search. Opinet publishes every station’s price on a public website, so a driver can compare prices across nearby stations at little cost. Where rivals are close, a station that delays a cut loses customers quickly; where the nearest rival is several kilometers away, the cost of delay is small. Search-based explanations of asymmetric adjustment predict that competition matters most for decreases, which is the pattern in Table 3: the 1-km gradient for the cut is twice that for the increase.

The second is inventory and turnover. Stations buy fuel in batches, and a station that holds inventory bought at the old supply price may wait to sell it before passing on a cut. Stations in dense markets and self-service stations sell more fuel and replenish more often, so they reach the new cost sooner. This explanation predicts faster adjustment in both directions, and it fits the larger coefficient for self-service stations and the fading of differences after two weeks, when even low-volume stations have turned over their stock. Without station-level sales I cannot separate it from search.

The third is attention. Each revision was announced publicly and covered by the press, and the government warned stations against raising prices unfairly. Drolsbach et al. (2023) attribute the full pass-through of the European tax cuts in part to public awareness and the threat of antitrust action. Monitoring and media attention are likely to be greater in dense urban markets, which would also speed cuts there. The equal timing of first price changes across market structures, within zero to two days everywhere, suggests that the difference lies less in noticing the revision than in how far each station moved once it did.

These explanations share an implication for policy. A revision of a wholesale cap is a public, uniform cost change, and the speed at which it reaches drivers varies with local market structure while the level reached does not. The evidence is an association: the number of nearby rivals is not randomly assigned, and the robustness check in Section 4.6 shows that part of the gradient reflects stations’ pre-existing pricing habits. Where rivals are few, the delay costs drivers a few won per liter for one to two weeks after each cut.

5.2 Policy implications

Timing of evaluations. A wholesale cap with announced revisions reached drivers quickly but not at once: more than half of each revision in the first week and most of it within three. Judging a revision by pump prices a few days later understates its effect. Evaluations of the cap, which

have focused on its effect on consumer prices—the government puts it at -0.6 percentage points on average over March–August (Ministry of Trade and Industry, 2026b)—should allow two to three weeks after each revision.

Where to monitor. Drivers in thin local markets received cuts later. Stations without a rival within 1 km had passed through 21 percentage points less of the June cut after one week than stations with six or more. Monitoring after a cut, of the kind the government announced against stations that raised prices unfairly (Ministry of Trade and Industry, 2026g), is most useful where stations face few nearby rivals. A count of rivals within 1 km, available from public station lists, is a rule of thumb of the kind Valadkhani et al. (2021) propose for Sydney.

Who bears the cost. A cap far below cost shifts its burden upstream. In Korea’s case the average retail margin was unchanged, and the gap between cost-based and capped supply prices—an opportunity cost of about 3.0 trillion won in April–August by my estimate, not an accounting loss—fell on refiners; the concurrent fuel-tax cut cost about 2.0 trillion won in forgone revenue, a separate budget cost. Compensating refiners on a cost basis, as the government chose to do, transfers much of the first burden to the budget as well. The longer a cap stays far below cost, the stronger refiners’ incentive to divert output to exports or draw down inventories, which the export limits were designed to contain.

Lessons beyond Korea. Korea’s choice of instrument was unusual. Of the 27 EU member states, more than half cut excise duty or VAT on gasoline or diesel between March and September 2026 (European Commission, 2026), and several, including Greece, Romania and Czechia, capped retail margins;¹ to my knowledge none capped refiners’ prices. The three instruments used since 2021 differ in who bears the cost and in how fast they work. Temporary fuel-tax cuts in Europe reached consumers largely, though not always fully, within days to weeks and cost the budget directly (Drolsbach et al., 2023; Kahl, 2024). China’s retail ceiling became a reference price that stations followed up faster than down (Cui et al., 2023), and Hungary’s was followed by a price surge after its removal (Berezvai and Helfrich, 2024). Korea’s wholesale cap reached consumers within two to three weeks without making the cap a focal retail price, but placed the burden on refiners until they are compensated. Governments considering a wholesale cap should expect a lag of weeks, slower relief in thin local markets, and compensation claims from refiners.

Competition enforcement. In July 2026 the four refiners were indicted over how they set supply prices. The evidence here cannot speak to that case. It does show that, while the cap lay at or below the cost-based price, refiners’ near-identical prices just below the cap are what any form of competition would produce, and that neighboring stations of the same brand did not time

¹Greek City Times, “Greece caps profit margins on fuel and essential goods,” 12 March 2026; Romania Insider on Emergency Ordinance 19/2026, March 2026; Euronews, “Czechia caps fuel margins and cuts diesel tax as prices soar,” 22 September 2026.

their changes together. Parallel pricing during a binding cap is weak evidence of coordination; authorities need other evidence, and the period after the cap is lifted deserves particular attention, as Hungary’s experience suggests (Berezvai and Helfrich, 2024).

6 Conclusion

Korea’s 2026 cap on refiners’ fuel prices offers a rare natural experiment: a wholesale price control that lay below refiners’ cost-based price throughout, revised on announced dates, with daily prices observed at every station. Each revision reached pump prices within two to three weeks, faster where stations had more nearby rivals, and the gap between cost and cap fell on refiners and the public budget rather than on station margins. Four limitations deserve emphasis. The paper measures how each revision reached pump prices, not how far the cap held prices below their level without it; because the cap applied to every station on the same date, no domestic control group exists for that question. The analysis covers one increase and one cut, so the asymmetry between them is descriptive. Refiners’ prices are weekly averages, so the introduction’s cost change and any within-week settlement are not observed. And without station-level sales, the incidence estimates rest on national shipments and a counterfactual pricing rule. As the cap continues to be revised, further revisions will allow the asymmetry to be tested as registered, and the eventual removal of the cap will show whether pass-through on the way out mirrors the pass-through documented here.

Declarations

Data availability. All data are public: Opinet (station and refiner prices, international product prices), the Bank of Korea’s ECOS (exchange rates), Petronet (shipments) and VWorld (geocoding). Code and the registration documents will be deposited on publication.

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Declaration of generative AI and AI-assisted technologies in the writing process. During the preparation of this work the author used an AI assistant (Anthropic’s Claude) to draft text and to write and run analysis code. The author reviewed, supervised and edited all text, code and results, and takes full responsibility for the content of the publication.

Competing interests. None.

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